Percentages, pricing & interest

10 practical calculations / formulas + worked examples

Use decimal rates: 12% = 0.12. Match rate and time periods. Multiply rate results by 100 for percent. Keep the same currency; round final results. Examples exclude taxes and fees.

01 Percentages

```
part = amount × p
p = part / amount
```

p = decimal percentage; amount must be nonzero in the second formula.

Example: 15% of 240 = 36; 36 / 240 = 15%.

Remember: 10% to 12% = 2 percentage points, or a 20% relative increase.

02 Percentage change

```
change = (new − old) / old
new = old × (1 + change)
```

Use a positive old value for ordinary price and balance comparisons.

Example: 80 to 100 = 25% increase. 100 to 80 = 20% decrease.

Remember: A zero starting value has no defined percentage change.

03 Successive discounts

```
final price = original price × (1 − d1) × (1 − d2)
combined discount = 1 − (1 − d1) × (1 − d2)
```

d1 and d2 = decimal discounts; each applies to the remaining price.

Example: 200, then 20% off, then 10% off = 144. Combined discount: 28%.

Remember: Successive discounts multiply. Adding 20% and 10% gives the wrong discount.

04 Margin vs. markup

```
profit = selling price − cost
margin = profit / selling price
markup = profit / cost
```

Margin is based on selling price. Markup is based on cost.

Example: Cost 80, price 100, profit 20: margin 20%; markup 25%.

Remember: Use the same tax basis. Target price for margin m: cost / (1 - m), m < 1.

05 Simple interest

```
interest = P × r × t
final balance = P + interest
```

P = principal; r = annual decimal rate; t = years.

Example: 1,000 at 12% a year for 6 months: interest 60; final balance 1,060.

Remember: Interest is on original principal. For days, use the contract's day-count basis.

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Sources: OpenStax, Contemporary Mathematics, sections 6.1-6.3 (openstax.org).

Growth, returns & borrowing

10 practical calculations / formulas + worked examples

Use decimal rates: 12% = 0.12. Match rate and time periods. Multiply rate results by 100 for percent. Keep the same currency; round final results. Examples exclude taxes and fees.

06 ## Compound interest

```
FV = P × (1 + r/m)^(m × t)
interest earned = FV − P
```

P = principal; r = nominal annual rate; m = compounds/year; t = years.

Example: 1,000 at nominal 12%, monthly for 1 year: final 1,126.83; interest 126.83.

Remember: Fixed rate, whole compounding periods, no deposits or withdrawals.

07 ## Effective annual rate

```
EAR = (1 + r/m)^m − 1
periodic rate from EAR = (1 + EAR)^(1/m) − 1
```

r = nominal annual decimal rate; m = compounding periods per year.

Example: Nominal 12%, compounded monthly = 12.68% effective annually.

Remember: EAR here excludes fees. Advertised APR and total-cost measures can differ.

08 ## Compound annual growth rate (CAGR)

```
CAGR = (ending value / starting value)^(1/t) − 1
```

t = years; positive starting and ending values, with no extra cash flows.

Example: 10,000 to 14,641 in 4 years = 10% CAGR.

Remember: CAGR is a smoothed rate. It does not describe each year's actual return.

09 ## Inflation-adjusted return

```
real return = (1 + nominal return) / (1 + inflation) − 1
```

Nominal return and inflation must cover the same period; both > -1.

Example: 15% nominal return and 10% inflation: 1.15 / 1.10 - 1 = about 4.55%.

Remember: Subtracting rates gives an approximation. Use comparable, decimal rates.

10 ## Fixed loan payment

```
payment = P × i / (1 − (1 + i)^(−N))
if i = 0: payment = P / N
```

P = principal; i = decimal rate per payment period; N = number of payments.

Example: 10,000 over 12 months at nominal 12%/year: about 888.49/month; at 0%: 833.33.

Remember: End-of-period payments, zero final balance. Final payment may adjust for rounding.

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Sources: OpenStax, Contemporary Mathematics, section 6.4 and Formula Review; St. Louis Fed, FRED Blog (fred.stlouisfed.org).